

Proposed Use of Engineering Time: Rate-Matched Imaging of the Geosynchronous Belt

C. Stubbs, on behalf of the GEO luminosity-function team

August 14, 2026

1 Project goal

The science objective is a **deep luminosity function of objects in and around the geosynchronous (GEO) belt** — active satellites, dead payloads, and debris — extending well below the completeness limit of current catalogs. Depth requires long effective integrations; long integrations on GEO objects require tracking that *freezes the targets, not the stars*.

2 Engineering-time objectives

1. Obtain **representative frames containing known artificial satellites** across a range of orbital inclinations, to exercise and hone the data-reduction pipeline (streaked-star astrometry, target photometry, co-addition).
2. **Validate and refine the observing strategy** described below — in particular the inclination-dependent rate-lock budget of §5, which fixes the exposure time, sequence length, and the readout overhead we can tolerate.

3 Observing strategy

We will **rate-match in right ascension as a function of inclination**: the telescope tracks at the apparent RA rate of a geosynchronous object of the chosen inclination class, freezing satellites and debris of that class in the image. Stars streak; we have previously demonstrated the ability to extract **reliable centroids and precision astrometry from stellar streaks**, so the astrometric solution is not degraded.

Observations are scheduled at each target’s **southern declination peak** (the culmination of its daily analemma arc), for three reasons: (i) the declination rate passes through zero there, so a pure-RA rate match is exact at mid-sequence; (ii) the RA rate is itself at an extremum there (§5), so it holds constant to high accuracy; and (iii) the southern turning point maximizes angular separation from the crowded zero-inclination belt.

At each tracking rate we will take a **succession of short exposures for co-addition, totalling** $T_{\text{int}} = 200 \text{ s}$ of integration. Field centers are offset north so the target sits 75% of the way toward the *southern* edge of the field of view at culmination; the object then climbs back north through the field, maximizing dwell time. Nightly target lists (LMST-ordered peak times, pointings in RA/Dec/HA/Alt/Az, and tracking rates with signs defined) are generated by the accompanying `geo_peaks.py` planning tool from same-day TLEs.

Sign conventions. RA rate > 0 means RA increasing (eastward motion). The sidereal reference rate is $\dot{\alpha}_{\text{sid}} = 15.0411''/\text{s}$ (the LMST rate, i.e. the RA rate of an ideal zero-inclination geostationary point). We quote object rates and the offset $\Delta \equiv \dot{\alpha} - \dot{\alpha}_{\text{sid}}$; $\Delta > 0$ means the object drifts *east* of a geostationary point. At the southern peak an inclined object runs eastward-fast: $\Delta \simeq f \dot{\alpha}_{\text{sid}} (\sec i - 1)$, where $f \simeq 1.17$ is the measured topocentric amplification from CTIO (Table 1).

Night thirds and illumination. The night (Sun below -12°) is split into thirds, and within each third targets are ranked by **solar phase angle** (Sun–object–observer; smaller = better lit). Because the anti-solar point rises in the east at dusk, crosses the meridian at midnight, and sets in the west at dawn, this ranking automatically points the first third **east**, the middle third near the **meridian**, and the last third **west**. The planner also flags objects that sit inside **Earth’s shadow** at their peak (a cylindrical-umbra test); eclipsed targets are excluded from the plan.

4 Passband selection

Rule: g band when the Moon is down; r band in bright time. Moonlight is scattered sunlight, and Rayleigh scattering scales as λ^{-4} : the lunar sky brightens g by $\gtrsim 2$ mag between dark and bright time, but r by only ~ 1 mag. In dark time the CTIO sky is faintest in g ($\mu_g \approx 22.0$ vs. $\mu_r \approx 21.0$ mag arcsec $^{-2}$), and g sits near peak system throughput, so g delivers the deepest limiting magnitude. Once the Moon is up, the g sky floods and r becomes the depth optimum. A secondary effect points the same way: satellite reflectance spectra (solar panels, MLI, aged coatings) are typically *reddened* relative to solar, so the targets themselves yield a small additional advantage in r that costs little in dark time and helps in bright time.

Two operational cautions. First, the crossover is governed by **Moon–target separation**, not Moon phase alone: a gibbous Moon 40° from the field damages g nearly as much as a full Moon 90° away. Second, the illumination scheduling of §3 prefers *anti-solar* pointings — and at full Moon the anti-solar point is *at the Moon*. In bright time we therefore back off from the lowest-phase-angle pointing as needed to hold $\gtrsim 30$ – 40° of lunar separation, accept a modestly larger phase angle, and observe in r . The engineering run will measure the actual g/r depth crossover directly (§6, item 5).

5 Rate-lock analysis: how long does the match hold?

For a circular geosynchronous orbit of inclination i observed at its southern peak (time t measured from culmination, $n = 2\pi/86164\text{s}$):

RA is not the constraint. The apparent RA rate is at an extremum at the peak; its residual after matching at $t = 0$ grows only cubically,

$$\delta\alpha(t) = \frac{n^3 \sin^2 i}{3 \cos^3 i} t^3 \quad \implies \quad \delta\alpha(\pm 300\text{s}) < 0.05'' \text{ even at } i = 14^\circ. \quad (1)$$

A single commanded RA rate therefore holds for the whole sequence.

Declination drift sets the budget. The declination rate is zero at the peak but accelerates away from it:

$$\ddot{\delta} = k(i) = f i n^2 \quad (\text{topocentric factor } f \simeq 1.17 \text{ from CTIO, measured}). \quad (2)$$

We verified k numerically with current TLEs propagated topocentrically from CTIO (Table 1); the measured values match $f i n^2$ to a few percent. An exposure of length τ taken at time t from the peak smears the target in declination by $k |t| \tau$. Requiring per-exposure smear below a budget ϵ confines the sequence to the window

$$|t| \leq \frac{\epsilon}{k(i) \tau} \quad \implies \quad W = \frac{2\epsilon}{k(i) \tau} \quad (\text{total usable wall time, centered on the peak}). \quad (3)$$

With $N = T_{\text{int}}/\tau$ exposures the sequence needs wall time $N(\tau + R)$ for readout time R , so the **maximum tolerable readout** is

$$R_{\text{max}} = \frac{W}{N} - \tau = \frac{2\epsilon}{k(i) T_{\text{int}}} \tau - \tau. \quad (4)$$

Table 1: Measured declination acceleration k at the southern peak, from current TLEs propagated topocentrically from CTIO (2026-08-14). The implied topocentric factor $f = k/(i n^2)$ is constant at $\simeq 1.17$.

Object	i (deg)	k ($'' \text{s}^{-2}$)	implied f
ANIK F2	3.43	7.5×10^{-5}	1.15
GALAXY 3C	3.88	8.7×10^{-5}	1.17
DIRECTV 5	5.02	1.10×10^{-4}	1.14
INTELSAT 902	6.26	1.40×10^{-4}	1.17
INTELSAT 904	7.06	1.54×10^{-4}	1.14
NIMIQ 2	9.58	2.09×10^{-4}	1.14
TDRS 3	12.56	2.81×10^{-4}	1.17
USA 115	14.09	3.18×10^{-4}	1.18
TDRS 6	14.18	3.19×10^{-4}	1.17

Table 2: Rate-lock budget for $T_{\text{int}} = 200$ s: usable window W and maximum tolerable readout R_{max} (Eq. 4), for per-exposure declination smear budget ϵ and exposure time τ . Negative R_{max} means 200 s of integration *cannot* be completed in a single peak visit at that configuration.

i (deg)	k ($''/\text{s}^2$)	$\epsilon = 0.5'', \tau = 20$ s		$\epsilon = 0.5'', \tau = 10$ s		$\epsilon = 1.0'', \tau = 20$ s	
		W (s)	R_{max} (s)	W (s)	R_{max} (s)	W (s)	R_{max} (s)
3	6.7×10^{-5}	744	+54	1488	+64	1488	+129
5	1.1×10^{-4}	446	+25	893	+35	893	+69
8	1.8×10^{-4}	279	+8	558	+18	558	+36
10	2.2×10^{-4}	223	+2	446	+12	446	+25
12	2.7×10^{-4}	186	−1	372	+9	372	+17
14	3.1×10^{-4}	159	−4	319	+6	319	+12

Conclusions from the budget.

- **Low inclination** ($i \lesssim 5^\circ$): 20-s exposures with generous readout ($R \lesssim 25$ s) comfortably reach 200 s in one visit. Readout losses are not a driver here.

- **Mid inclination** ($i \sim 8\text{--}10^\circ$): 20-s exposures work only if readout is fast ($R \lesssim 2\text{--}8$ s). With a DECam-class readout ($R \approx 20$ s), *shorter exposures are required*: at $\tau = 10$ s the budget allows $R_{\text{max}} \approx 12\text{--}18$ s — marginal. This is precisely the regime the engineering run must probe.
- **High inclination** ($i \gtrsim 12^\circ$): a single peak visit cannot deliver 200 s at $\epsilon = 0.5''$, $\tau = 20$ s at all ($W < 200$ s). Options, in order of preference: (i) **command a per-frame declination rate** ($\dot{\delta} = kt$, a linear ramp): the residual per exposure drops to $\frac{1}{2}k\tau^2 \lesssim 0.06''$ and the window is then limited only by the cubic RA term, i.e. many tens of minutes; (ii) shorten exposures to $\tau = 10$ s; (iii) relax the smear budget to $1''$; (iv) split the sequence across two consecutive peak passages and co-add across visits.
- Because the drift is *symmetric* about the peak, sequences must be **centered on the peak time** (start $W/2$ before culmination) — the LMST-ordered peak times from `geo_peaks.py` are the sequence midpoints.
- If co-addition registers each frame on the measured target position (shift-and-add), the accumulated declination *offset* across the sequence (up to $\sim \frac{1}{2}k(W/2)^2 \approx 3\text{--}13''$) is removed and only the per-exposure smear above matters; if instead fixed-WCS stacking is used to preserve star-streak astrometry, the offset itself must fit the smear budget and the windows above shrink by $\sqrt{2\epsilon/(kW\tau)}$. Testing both reduction paths is an explicit engineering goal.

6 Proposed engineering sequences

1. **Rate ladder** (one visit each): known satellites at $i \approx \{0, 3.5, 5, 7, 9.6, 12.6, 14.2\}^\circ$ from the nightly `geo_peaks` list (e.g. EchoStar 10, ANIK F2, DIRECTV 5, INTELSAT 904, NIMIQ 2, TDRS 3, TDRS 6), each with a 200-s co-add sequence centered on its southern peak, at the matched RA rate and $\dot{\delta} = 0$. Verifies the rate-lock windows of Table 2 directly against trailed-image metrology.
2. **Declination-ramp test** at $i \approx 12\text{--}14^\circ$: repeat with the per-frame $\dot{\delta}$ ramp, to qualify option (i) above and unlock the high-inclination population for the science survey.
3. **Exposure-time pair** at $i \approx 10^\circ$: $\tau = 10$ s vs $\tau = 20$ s sequences of equal T_{int} , to measure the real readout overhead and the smear–depth tradeoff in the pipeline.
4. **Pipeline fields**: two zero-rate-offset fields on the crowded zero-inclination belt (streaked stars, frozen belt objects) for the luminosity-function reduction chain end-to-end.
5. **g/r depth crossover** (§4): one field observed in both g and r , in dark time and again with the Moon up at two lunar separations ($\sim 40^\circ$ and $\sim 90^\circ$), with equal T_{int} — fixing empirically where the band switch belongs and quantifying the reddened-target advantage in r .

Total request: the sequences above fit comfortably in **two half-nights** of engineering time, including calibrations, with peak times spread across the LMST range (the nightly list spans LMST $\approx 15\text{--}04$ h for the inclined fleet visible from CTIO).

Tooling

`geo_peaks.py` (this directory) generates the nightly list from same-day TLEs (CelesTrak or IAU CPS SatChecker): LMST-ordered southern-peak times, object and pointing RA/Dec, hour angle, alt/az, RA rates and offsets from sidereal with the sign conventions above, zenith angles, and sub-satellite longitudes, plus declination-track, sky-position, and rate-offset plots. All numbers in §5 were computed with the same propagation code used for the nightly lists.